The route architecture, on one page

Compare direct IPC paths and switch expansion first; display and alarms form a shared output layer.

COMMON SIGNAL LOGIC



FIVE ENGINEERING ROUTES

Route	Main unit / geography	IPC camera count	Connection	Switch / procurement difference
C06	M3N H0404 · North America	4 × C46	**4 direct IPC paths**	No switch; HDD/SSD not in kit
C10	M3N H0204 · EU/Asia	4 × C46	**4 direct IPC paths**	No switch; 1 TB SSD in kit
C07	M1N 2.0 · all regions	2 × C46	**2 direct IPC paths**	No external switch
C08	M1N 2.0 · all regions	4 × C46	**4 via switch**	Switch ×1 + drive cable ×1
C09	M1N 2.0 · all regions	5 × C46	**1 direct + 4 switched**	Same switch + drive cable

C01-C05 and C11 are reference applications, not ready-to-order construction kits; see the concept-reference section (page 11).

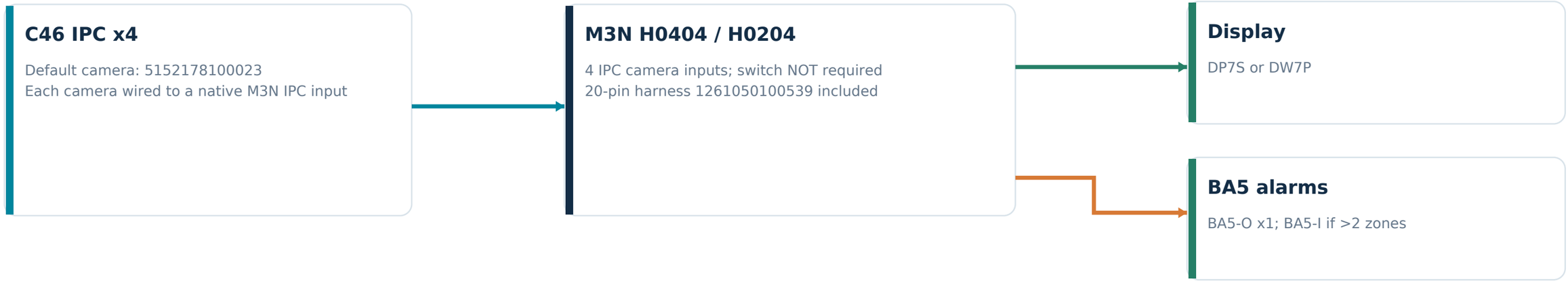
Source: v4 stage-2 routes + current confirmed BOM rules (2026-10-22)

M3N | 4 IPC cameras, direct

H0404 and H0204 share the same four-IPC architecture; regional kit contents differ.

NEXT: TWO M3N DRAWINGS

4 direct IPC connections



SELECT THE REGIONAL HOST KIT

C06 | H0404 · North America

NA STANDARD

5120121100005

HDD enclosure INCLUDED, actual HDD/SSD NOT included. 20-pin, power cable and regional antennas are in-kit.

C10 | H0204 · EU / Asia

EU / ASIA ONLY

5120182100013

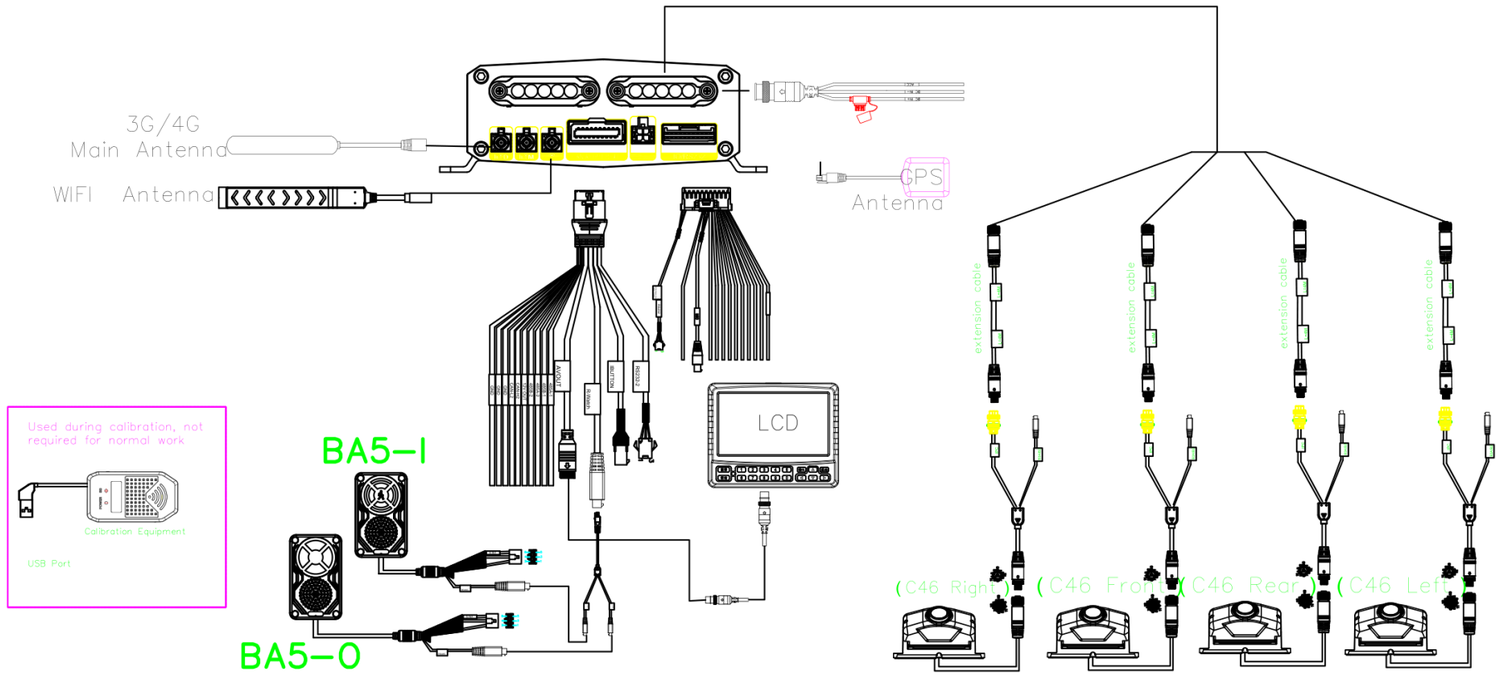
Includes 1 TB SSD, SSD enclosure, GPS / 4G / Wi-Fi and 20-pin cable. No MicroSD card. Not a standard NA kit.

M3N source drawing contains legacy 22-pin depictions; use confirmed current 20-pin kit, not old harness 1261050100409.

ENGINEERING CONNECTION / CURRENT ROUTE MAPPING

C06 / C10 | M3N Four-Camera Reference

4 direct IPC camera paths | H0404 North America / H0204 EU-Asia



Source A4 | C06 / C10

01 Direct IPC topology

Four C46 IPC units use the four native M3N camera paths. No external switch is required.

02 Host kit by region

C06: H0404 NA 5120121100005 (enclosure only; no drive). C10: H0204 EU/Asia 5120182100013 (includes 1TB SSD).

03 Current 20-pin cable

Both current host kits include 1261050100539. The source shows a legacy 22-pin depiction; DO NOT order or infer pinouts from it.

04 BA5 wiring notes

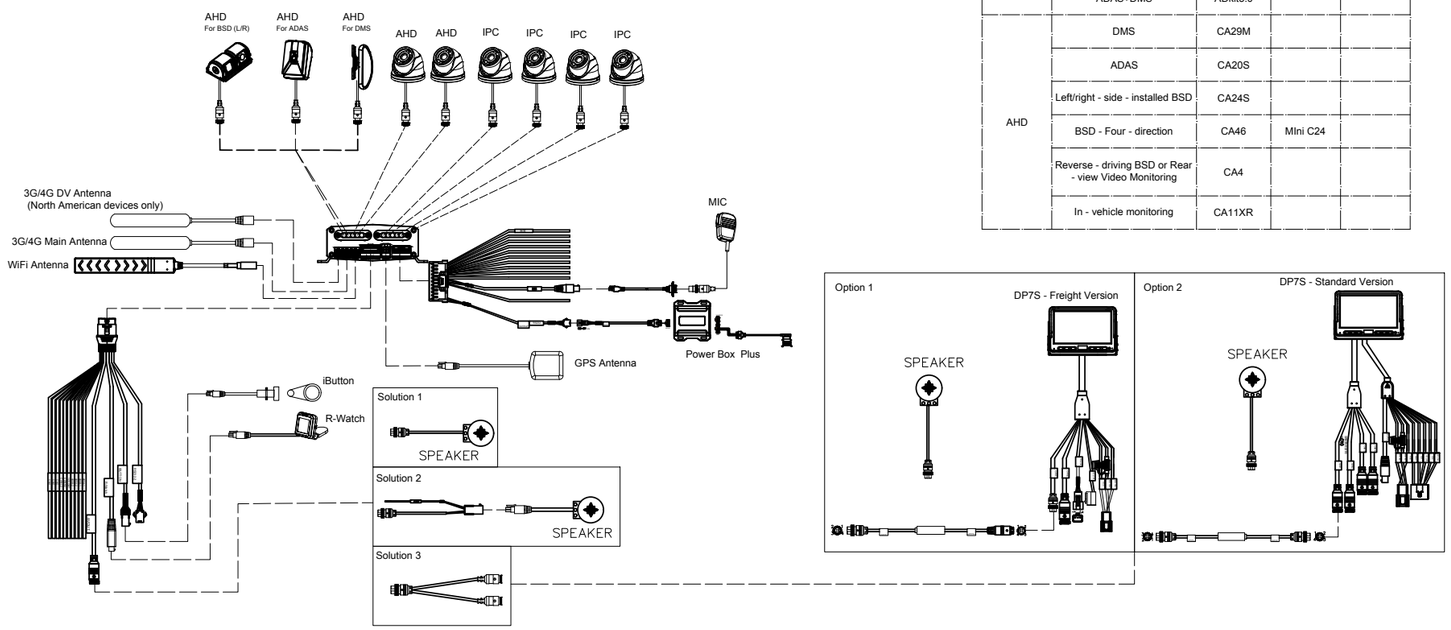
BA5 shown here is conceptual. Current dual BA5-I/O uses M12-4 0004 variants and adapter 1261050100490 ×1.

Shared architecture reference only—not the final 20-pin pinout for either H0404 or H0204.

Source: M3N+C46-□□.pdf

M3N | General System Connection Reference

Ports / expansion options | Not the procurement wiring sheet for C06 or C10



Camera expansion options (M3N supports 4 AHD and 4 IPC)				
IPC	DMS	C29N		
	ADAS+DMS	ADkit3.0		
AHD	DMS	CA29M		
	ADAS	CA20S		
	Left/right - side - installed BSD	CA24S		
	BSD - Four - direction	CA46	MIni C24	
	Reverse - driving BSD or Rear - view Video Monitoring	CA4		
	In - vehicle monitoring	CA11XR		

Source A5 | General M3N

01 Expansion capability

The generic diagram illustrates 4 AHD + 4 IPC options, including DMS / ADAS / BSD camera families.

02 Current Construction route

C06/C10 only select the four-C46-IPC architecture. This does not imply all additional functions are included.

03 Display and accessories

Legacy DP7S display and audio options are shown. Apply current display and BA5 ordering / compatibility rules.

04 Legacy harness warning

The 22-pin Adapter Cable on this generic source is NOT the current 20-pin 1261050100539 and does not establish pin equivalence.

Reference for general M3N capability, not a released complete H0404/H0204 kit or pinout.

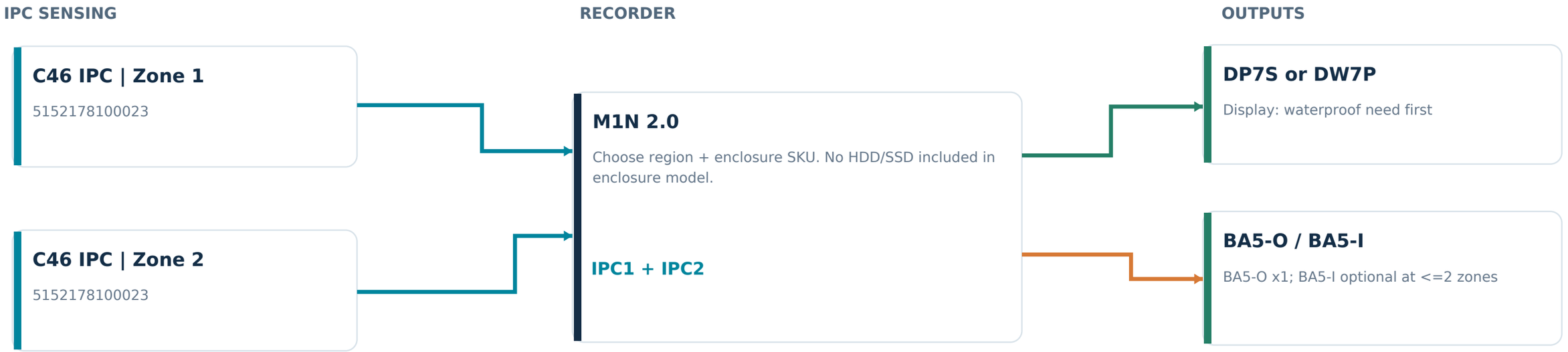
Source: M3N System Connection Diagram- V1.1-20260104.pdf

M1N 2.0 | 2 IPC cameras

Two cameras use the native IPC connections of the recorder; no external switch is involved.

NEXT: C07 NATIVE WIRING

No switch



PER-VEHICLE DIFFERENCE

Host

6 regional M1N SKUs (enclosure / no enclosure)

Camera cabling

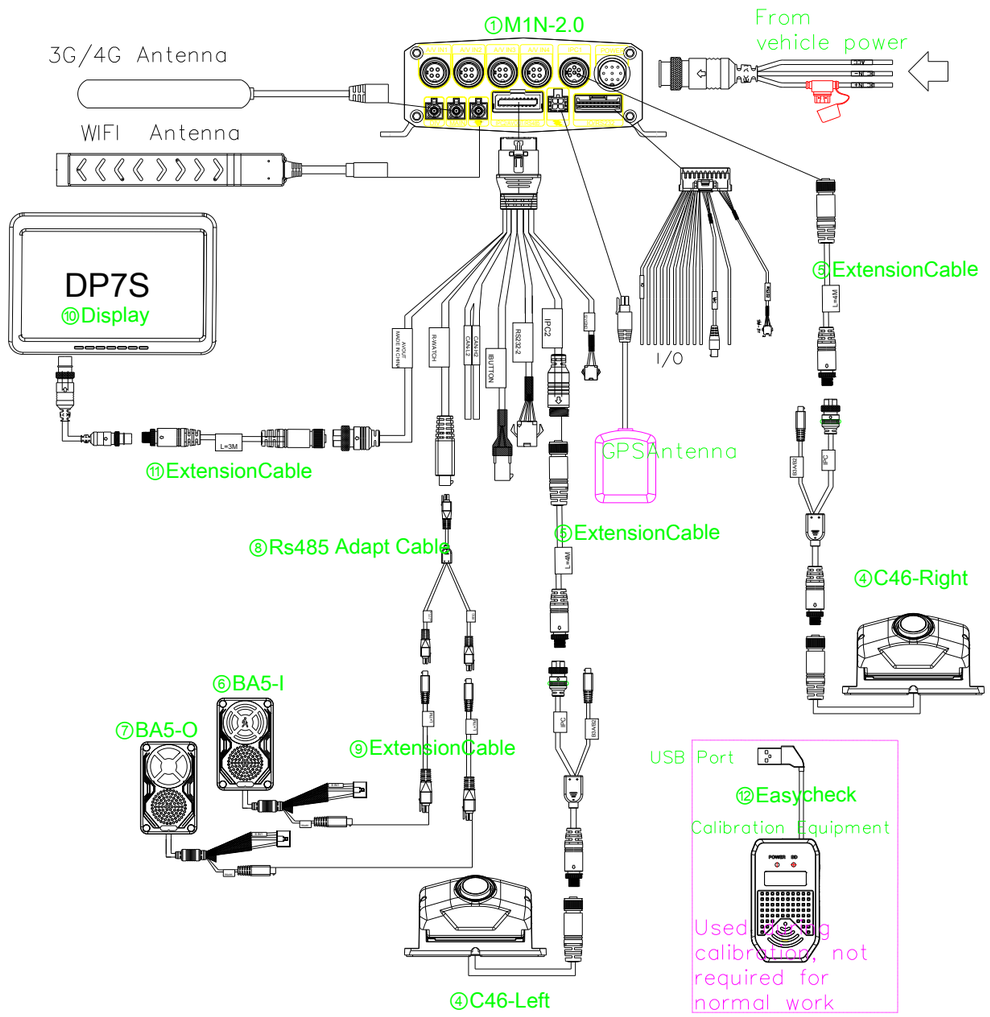
RS765-6 extensions by run length; 3 / 5 / 7 / 11 / 15 m

Alarm kit

If both BA5 are fitted: 1261050100490 x1 separately

C07 | M1N Two-Camera Native Wiring

2 direct C46 IPC paths into M1N | No external switch



Source: M1N-2.0+2C46+BA5.pdf

Source A1 | Route C07

01 IPC camera routing

Two C46 IPC cameras (5152178100023) use the native M1N IPC paths; no external switch.

02 Display and alarms

Select DP7S / DW7P based on waterproof requirement. BA5-O ×1 is standard; BA5-I is optional for this two-zone example.

03 Dual-alarm adapter

When BA5-I and BA5-O are both fitted, separately order RS485 adapter 1261050100490 ×1 per vehicle.

04 Service / cable lengths

Easycheck is a service tool, not permanently connected. L=4M / L=7M on the native drawing are illustrative, not current orderable SKUs.

Source is a connection reference. Current BA5 procurement uses 0004 versions; perform second BOM verification.

M1N 2.0 | 4 vs 5 IPC cameras

One external switch is required in either route. Five cameras do NOT all pass through the switch.

NEXT: C08 / C09 DRAWINGS

C08 | Four cameras

1 switch

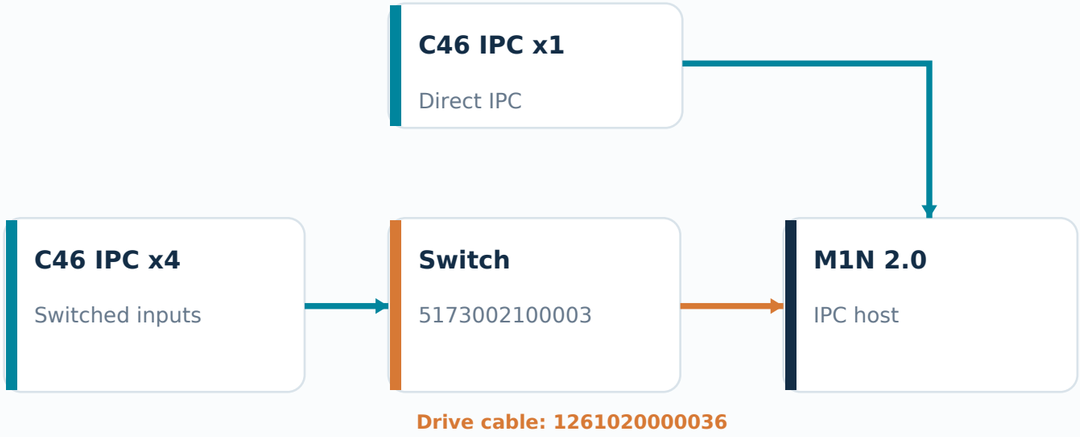


4 CAMERA CONNECTIONS

One switch and one drive cable. No separate switch power accessory.

C09 | Five cameras

1 switch



1 DIRECT + 4 VIA SWITCH

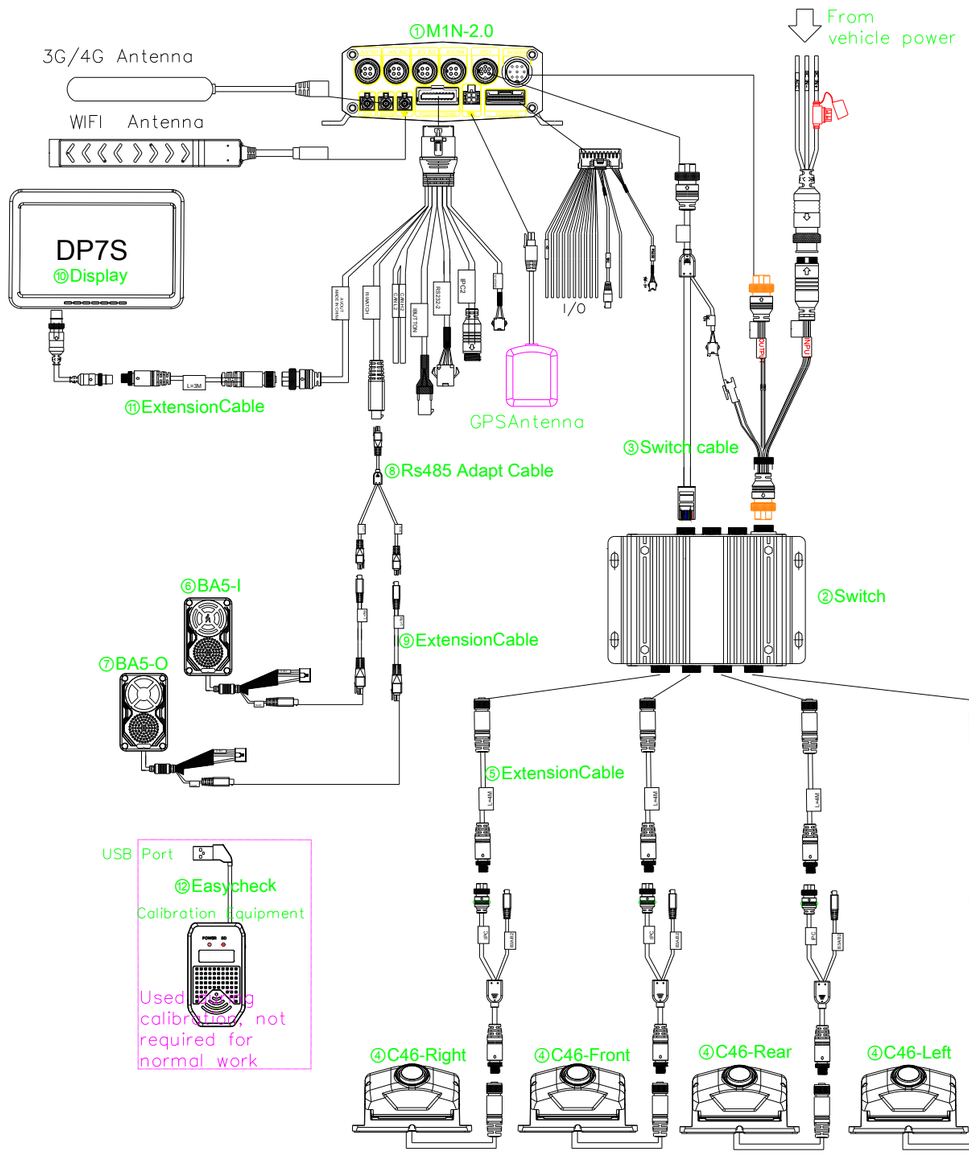
One switch and one drive cable. No separate switch power accessory.

Shared outputs (not redrawn): display DP7S / DW7P, BA5-O x1 and BA5-I recommended for >2 zones. When both alarms are fitted, adapter 1261050100490 x1 is a separate order.

Sources: M1N-2.0+C46.pdf (4-camera) and M1N-2.0+5C46.pdf (5-camera)

C08 | M1N Four-Camera Native Wiring

All four C46 IPC channels enter M1N through the external switch



Source: M1N-2.0+C46.pdf

Source A2 | Route C08

01 Four switched paths

4 × C46 IPC → switch → M1N. Different from the direct two-camera route.

02 Mandatory network kit

Switch 5173002100003 ×1 + drive cable 1261020000036 ×1. No additional separately ordered switch-power part.

03 Operator outputs

Select DP7S / DW7P. BA5-O ×1 is standard; strongly recommend BA5-I ×1 for >2 blind-spot zones; dual alarms need 0490 ×1.

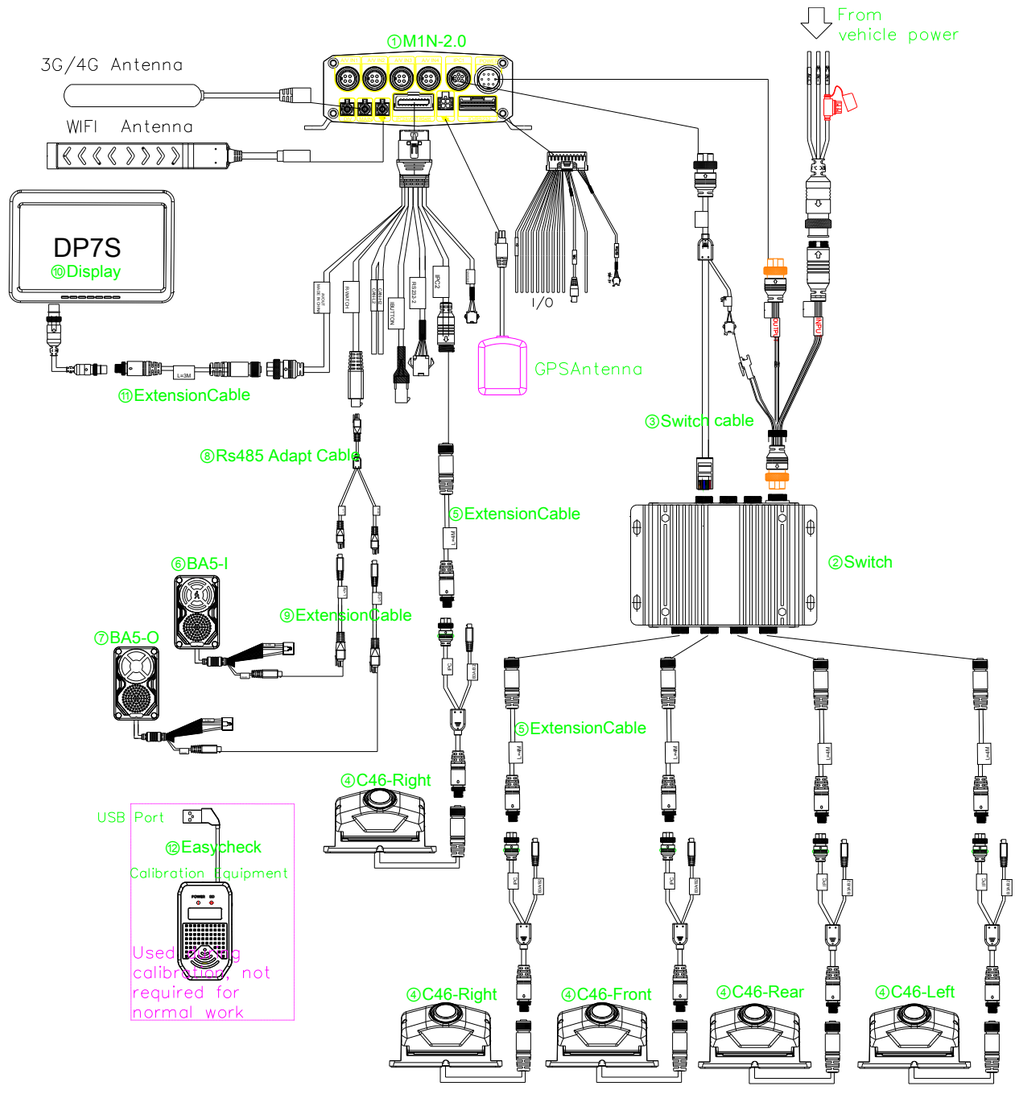
04 Camera extension note

The 4 m leads in the native drawing are schematic. Current lengths: 3 / 5 / 7 / 11 / 15 m; standard 7 m = 1260010000353.

Use current released wiring guidance for pinouts, power and vehicle IO, not the simplified native picture alone.

C09 | M1N Five-Camera Native Wiring

Key: 1 native direct IPC path plus 4 switched IPC paths



Source: M1N-2.0+5C46.pdf

Source A3 | Route C09

01 One direct branch

Camera 5 is connected to the native M1N IPC path, bypassing the switch.

02 Four switched branches

The other four C46 units go through switch 5173002100003 ×1 and drive cable 1261020000036 ×1.

03 Direction-label caveat

The source drawing labels two cameras as C46-Right. This appears duplicated; confirm the fifth installation zone against the actual vehicle.

04 Shared display / alarms

Display, BA5 and service tools use the common route rules. Easycheck is temporary commissioning equipment, not an onboard item.

Five cameras are NOT all connected through the switch. Preserve the direct + switched split.

Component decisions | What changes the BOM

Three decisions account for most configuration mistakes: camera connector, waterproof display and alarm combination.

01

IPC CAMERA

C46 0023 | Standard

5152178100023 · RS765-6 direct, non-waterproof connector. Standard bracket/screws included.

C46 0017 | Conditional

5152178100017 · waterproof M12 camera port; 1261010100072 adapter in kit. C06: PoC fallback only.

02

DRIVER DISPLAY

DP7S | No waterproof need

With-logo 5146026100001: MDVR power. No-logo 5190012100084: MDVR or vehicle power. Cable 1210030000163 separate.

DW7P | Waterproof required

5146054100001: MDVR or vehicle power; mount/base cable in kit. 3 m extension 1260010100265 separate.

03

ALARM COMBINATION

BA5-O | Standard

5164003100004 x1/vehicle. Alone: direct host harness; no dual-alarm adapter.

BA5-I | Strong recommendation

5164002100004 x1 if >2 zones; 4-direction indicator. With BA5-O order 1261050100490 x1 separately.

Camera RS765-6 extension lengths: 3 / 5 / 7 / 11 / 15 m. Standard 7 m = 1260010000353. BA5/DW7P 3 m waterproof extension 1260010100265 is shared.

Other applications | Concept references only

The original Stage-1 scenarios are consolidated; these are opportunities to scope, not fixed Construction BOMs.

Local viewing / alarms

Construction sites · Forklift / warehouse · Agriculture (C46A)

C46 AHD / local camera → **Vehicle interface** → Local display / alarm

Confirm power, triggers, display signal and protection before quoting.

Connected fleet / BSIS

On-road fleets · C46 IPC / C53 BSIS

Camera / C53 → **MDVR** → FT Cloud

C53 needs front mounting to monitor same-side long side and front blind spot. Regulation evidence is project-specific.

360° driver visibility

Premium trucks · AI-AVM / bird's-eye view

Surround cameras → **AI-AVM processor** → Driver 360° display

Driver-led visual checks before movement or low-speed turns; no blanket AI claim.

OEM / digital mirror

CM31 / CMS + C53 / DP12S (by OEM project)

OEM camera → **OEM interface** → CMS display

Mirror-replacement compliance requires vehicle / region approval.

Stage-1 concepts are intentionally condensed, rather than repeated as eight nearly identical full-page blocks.

Engineering handoff | Verify before release

Sales uses simplified architecture; FAE uses native drawings for power, pinouts and actual vehicle installation.

A clear four-step route from architecture to customer kit

01 Select route

Camera count + host region + switch need

02 Build full BOM

Cables, display, alarms, in-kit vs separate order

03 Re-verify SKUs

ERP/PDM check OR Product Manager confirmation

04 Release to customer

Only after kit completeness and version sign-off

DRAWING TRACEABILITY | USE THE NATIVE FILE FOR PINOUTS

Route	Source drawing	Use / constraint
C06/C10	M3N + C46 (model source)	4 direct IPC | page 3
REF	M3N System Connection Diagram- V1.1-20260104.pdf	General M3N | page 4
C07	M1N-2.0+2C46+BA5.pdf	2 direct IPC | page 6
C08	M1N-2.0+C46.pdf	4 via switch | page 8
C09	M1N-2.0+5C46.pdf	1 direct + 4 switch | page 9

The old 7 m cable 1260010100168 and old H0204 20-pin 1261050100409 in v4 are NOT current BOM defaults. Use 1260010000353 (7 m) and included 1261050100539 (20-pin).